# The Biomechanics and Physiology of Eliud Kipchoge's Marathon Records

Biokinesiology Project Report

## 1. Introduction and Research Question

Eliud Kipchoge is widely regarded as the greatest marathoner in history. He ran 2:01:39 in Berlin in 2018 (a world record at the time), 1:59:40 in the INEOS 1:59 Challenge in Vienna in 2019 (the first sub-two-hour marathon, though not record-eligible), and 2:01:09 in Berlin in 2022, another world record at the time. Kelvin Kiptum later lowered the official record in 2023.

These times demand extraordinary speed. In Vienna, 42,195 m in 7,180 s is about 5.88 m/s, or roughly 21.2 km/h (4:34 per mile). Berlin 2022 was about 5.80 m/s, or 20.9 km/h.

**Research question:** What biomechanical, physiological and neuromuscular factors allow an athlete to hold about 21 km/h for more than two hours?

## 2. Biomechanics of Running Gait

Running is a repeating gait cycle made of a stance phase, when the foot is on the ground, and a swing phase, when the leg recovers through the air. At marathon pace there is also a short flight phase when neither foot touches the ground.

**Stance phase.** At initial contact the foot lands close beneath the body, typically on the midfoot or forefoot for elite runners. During loading response and mid-stance the knee flexes to absorb impact while the quadriceps work eccentrically. At push-off the hip, knee and ankle extend together, with the gluteus maximus, quadriceps, gastrocnemius and soleus producing propulsion. Ground contact at this speed lasts only around 0.2 seconds.

**Swing phase.** The hip flexors (iliopsoas, rectus femoris) swing the thigh forward, the hamstrings flex the knee to bring the heel up toward the buttocks, and the leg then extends forward again. The tibialis anterior dorsiflexes the foot so the toes clear the ground, and the hamstrings and glutes decelerate the leg late in swing to prepare for contact.

**Cadence and stride length.** Elite marathoners typically run about 180 steps per minute or higher. Step length equals speed divided by step rate. At 5.88 m/s and 3 steps per second, step length is about 1.96 m. Long, slender legs and efficient mechanics help produce this without excessive effort.

**Posture and arms.** A slight forward lean from the ankles, relaxed shoulders, arms bent near 90 degrees swinging front to back, and minimal vertical oscillation all reduce wasted energy.

| Joint | Action | Key muscles |
|---|---|---|
| Hip | Extension / flexion | Gluteus maximus, hamstrings / iliopsoas, rectus femoris |
| Knee | Extension / flexion | Quadriceps / hamstrings |
| Ankle | Plantarflexion | Gastrocnemius, soleus |
| Ankle | Dorsiflexion | Tibialis anterior |
| Trunk | Stabilization | Transverse abdominis, obliques, erector spinae |

**Key concepts.** Ground reaction force (GRF) is the force the ground exerts back on the runner, typically around 2.5 to 3 times body weight at peak in distance running. The stretch-shortening cycle lets the Achilles tendon and arch of the foot store elastic energy during loading and return it during push-off, reducing the muscular work required. Running economy is the oxygen cost of running at a given speed, and it is one of the strongest differentiators among elite runners.

## 3. Exercise Physiology

**Energy systems.** The marathon is overwhelmingly aerobic. The phosphagen system and anaerobic glycolysis supply energy quickly but only for seconds to a couple of minutes, so at sustained marathon pace oxidative metabolism in mitochondria provides nearly all ATP.

**Determinants of performance.** The classic model (Joyner and Coyle, 2008) identifies three factors: maximal oxygen uptake ($VO_2$max), the fraction of $VO_2$max that can be sustained (linked to lactate threshold), and running economy. Kipchoge's advantage lies especially in the second and third.

**Muscle fibers.** Elite endurance athletes tend to have a high proportion of Type I (slow-twitch) fibers, which are fatigue-resistant, rich in mitochondria and capillaries, and efficient at oxidizing fat and carbohydrate.

**Fuel use.** At marathon intensity the body burns a mix of carbohydrate and fat, with carbohydrate dominating at faster paces. Glycogen stores in muscle and liver are limited, so depletion causes the sudden slowing known as "hitting the wall." Taking in carbohydrate during the race, commonly 60 to 90 g per hour, helps delay this. A rough energy estimate is about 1 kcal per kg of body mass per km, so a roughly 52 kg runner uses about 2,200 kcal over 42.195 km.

**Thermoregulation.** Running generates large amounts of heat. Sweating is the main cooling mechanism, and cool, stable weather keeps core temperature and cardiovascular strain lower. Fluid replacement limits dehydration.

**Body composition.** Kipchoge is lean and light (about 52 to 57 kg at around 167 cm). Lower body mass reduces the energy needed to move the body, and long, slender lower limbs have low rotational inertia, making each swing cheaper.

## 4. The Vienna 2019 Factors

The 1:59:40 run was not eligible as a record because it used conditions that go beyond standard competition rules:

- **Rotating pacemakers in a V formation** reduced air resistance on Kipchoge, which at 21 km/h is a meaningful share of the energy cost. Estimates of the saving are a few percent.
- **A laser pace guide** projected the target pace onto the road so he could hold an even speed without thinking.
- **A flat, looped course** with minimal turns and elevation change.
- **Advanced carbon-plate shoes** with thick foam improved running economy. Hoogkamer and colleagues (2018) reported an average improvement of about 4 percent over conventional racing shoes.

- **Optimized drinks delivery** from a support bike, so he did not need to slow down.

Together these show how even small percentage gains matter at the top level, and why the result remains an extraordinary demonstration of what is physiologically possible rather than an official record.

## 5. Training Principles

Kipchoge's training is reported to include around 200 km per week at peak, mostly at easy aerobic intensity with a smaller number of hard sessions such as tempo runs and intervals (a polarized approach). Each element drives specific adaptations:

- **High-volume aerobic running** raises mitochondrial density, capillary density and stroke volume, and improves fat oxidation.
- **Tempo and interval work** raises lactate threshold and the speed that can be sustained.
- **Strength and core work** improves tendon stiffness, force production and trunk stability.
- **Recovery, sleep and nutrition** allow adaptation to occur.
- **Periodization and tapering** build fitness gradually and reduce fatigue before the race.

## 6. Injury Prevention and Biomechanical Risk

Over a marathon each leg lands roughly 10,000 to 11,000 times, each time absorbing a force of several times body weight. Common overuse injuries include patellofemoral pain, iliotibial band syndrome, Achilles tendinopathy, shin splints and stress fractures. They arise when repeated loading outpaces the tissue's ability to adapt. Efficient mechanics, a cadence that avoids overstriding, gradual training progression, strength work and adequate recovery all lower risk.

## 7. Data and Model Results

This project includes an interactive simulation and three open datasets (see the README). The race results below are official times; every other value is produced by a simplified model using published rules of thumb, not measurements of Kipchoge.

| Race | Date | Time | Speed (km/h) | Pace /km | Record-eligible |
|---|---|---|---|---|---|
| Berlin Marathon 2018 | 2018-09-16 | 2:01:39 | 20.81 | 2:53 | Yes |
| INEOS 1:59 Challenge | 2019-10-12 | 1:59:40 | 21.16 | 2:50 | No |
| Berlin Marathon 2022 | 2022-09-25 | 2:01:09 | 20.90 | 2:52 | Yes |

At Vienna pace the model gives a step length of about 1.94 m at 182 steps per minute, ground contact of about 203 ms, a peak ground force near 2.7 times body weight, and roughly 10,890 foot strikes per leg over the marathon. A recreational runner at 12 km/h has longer contact (245 ms) and a shorter step (1.10 m) at the same cadence, and spends far longer on the course loading the same tissues.

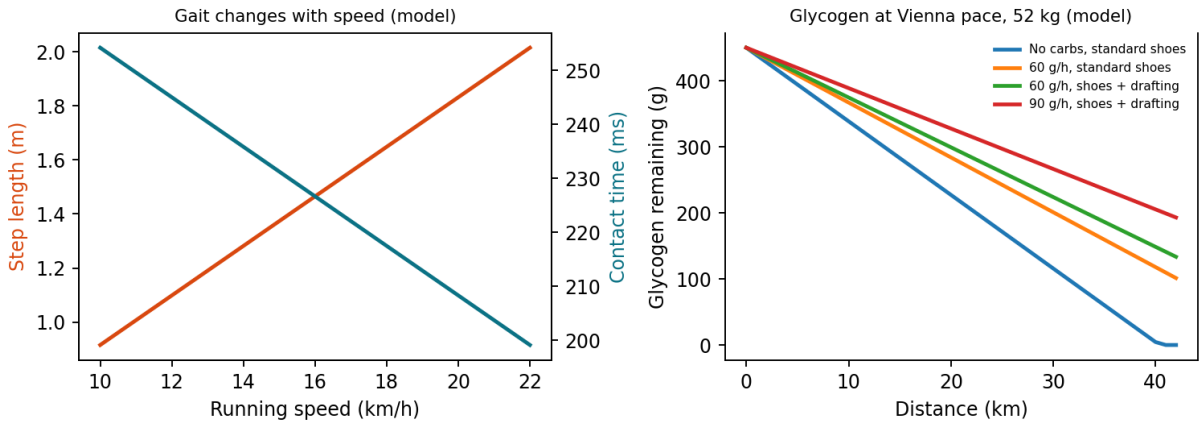



Figure 1. Left: model step length and contact time versus speed at 182 steps/min. Right: modeled glycogen remaining at Vienna pace for a 52 kg runner under four fueling and equipment scenarios. Without carbohydrate intake the 450 g store runs out near the end of the race, while fueling at 60 to 90 g/h keeps it positive.

**Practical component.** Choose one of: (1) video gait analysis in Kinovea at 120 fps, comparing cadence, foot strike and joint angles with Kipchoge footage; (2) a pace calculator comparing split times for 2:01:09 and 1:59:40 with your own 5K or 10K pace; (3) an energy cost estimate from body mass and distance, with a fueling plan.

## 8. Conclusion

Elite marathon performance is the product of several factors working together: efficient gait mechanics, a high aerobic capacity, outstanding running economy, a high sustainable fraction of $VO_2$max, disciplined training, careful fueling, and, in the case of Vienna, supporting technology and strategy. Kipchoge's records show how small advantages in each area add up to a performance that was once thought out of reach.

## 9. Sources


- Joyner, M.J. and Coyle, E.F. (2008). Endurance exercise performance: the physiology of champions. *Journal of Physiology*.
- Jones, A.M. et al. (2021). Physiological demands of running at 2-hour marathon race pace. *Journal of Applied Physiology*.
- Hoogkamer, W. et al. (2018). A comparison of the energetic cost of running in marathon racing shoes. *Sports Medicine*.
- World Athletics and INEOS 1:59 Challenge official materials.


Please verify citations and figures against the original sources before submitting.